

Supplementary Material: Diagnosing and Surgically Correcting Conservative Decision Bias in Imitation-Learned Air Combat Policies

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ABSTRACT

Supplementary material for the paper above. Section E records the provenance of every number quoted in the main text; nothing here is required in order to follow the argument.

KEYWORDS

reproducibility, provenance

ACM Reference Format:

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1 APPENDIX B. THE ENVIRONMENT'S LAUNCH MASK VERSUS THE EXPERT'S EXTRA GATE

The mask. The flat action mask is `[speed(3), heading(5), altitude(1), fire(2)]`. The launch branch is flat index `10.mask[10] = 1` requires *all* of:

The expert's gate. The rule expert never fires merely because `mask[10] = 1`. It additionally requires its own `fire_desired`, a pure function of the observation:

Why "strict subset" is exact, not rhetorical. With `r_max(AA) = 3000 + 5000 · (AA/180)`:

- lower bound: $1500 + 0.25 · (r_{\max} - 1500) > 1500$ whenever $r_{\max} > 1500$, i.e. for every $AA > 0$;
- upper bound: $1500 + 0.75 · (r_{\max} - 1500) = 0.75 · r_{\max} + 375 < r_{\max}$ whenever $r_{\max} > 1500$.

So on every axis the expert's admissible set is a strict subset of the environment's. This is the mechanical content of the paper's central claim that *the expert carries a conservative engagement preference that the environment does not require* — not a claim about the expert's competence.

Consequence to keep explicit: the mask, hence CLR's denominator, is policy-dependent. Firing triggers C1b, which *closes* the window; abstaining leaves it open. A policy that fires therefore collapses its own opportunity set, while a policy that abstains inflates it. Measured on the Figure-2 seed: **41** legal steps for BC versus **4** for SPC (a factor of 10.25; the aggregate in §4.2 is $\approx 10.8\times$). CLR is consequently a **descriptive** statistic across policies; the causal claim rests on the frozen-trajectory enumeration of §4.3, where the opportunity set is held fixed by construction.

2 APPENDIX C. THE ENUMERATED FIRE-POLICY FAMILY

All seven arms are evaluated on the *same frozen maneuver trajectory* per seed (the BC heading/speed actions), so only the launch decision varies (`scripts/fire_oracle_audit.py`):

`dlz_mid` and `dlz_deep` deliberately bracket the expert's own band `[0.25, 0.75]`, so the family contains a faithful approximation of the expert's gate as well as the mask-permissive extreme.

3 APPENDIX D. REPRODUCTION COMMANDS

All commands are for the **WSL Ubuntu shell**, from the repository root, using the project interpreter `/home/sean/miniconda3/envs/marl_env/bin/python`. Trace JSONs, logs and `.pth` checkpoints are excluded from version control, except the `E*.json` and `mechanism_*` artifacts, which are tracked.

Primary endpoints (§4.4, §4.5b, Table 4):

Frozen-trajectory enumeration (§4.3, Table 3):

Matched BC-versus-expert pair (§4.4, fresh env per episode — note this path also computes CLR as of 2026-09-17):

The ϵ -difficulty arms are the same two commands with `-difficulty 0.3`. The CLR measurement is the same command again with `--model-id paired_bc_vs_expert_d0_clr` and `--out results/shoot_eval/E3_paired_bc_vs_expert_d0_n400_s20000_v3_clr.json`.

Figures 1–3 and the auto-generated matrix:

`make_paper_figures.py` reads every plotted value from the tracked `E*.json` artifacts and holds **no numeric constants**, so Figures 1 and 3 cannot drift away from the text; if an artifact is missing the figure renders pending and warns rather than substituting a value from another basis. `audit_paper_numbers.py` recomputes the headline numbers straight from the artifacts and then checks that the draft still quotes them, including two must-be-absent guards: the abstract's expert-CLR placeholder (once the CLR run completes) and the bogus $2.5e-65$ p-value produced by feeding tie counts into `exact_mcnemar`. Both guards are themselves documented in the draft, so the checker ignores occurrences sitting inside a line that is explicitly talking about the trap.

`collect_matrix.py` reads **only** each file's `run_meta` — never filenames — and skips any artifact whose name is marked `EXCLUDED_`, so retired evidence cannot silently re-enter the table.

4 APPENDIX E. REPRODUCIBILITY AND ARTIFACT PROVENANCE

Supporting evidence for claims made in the main text. The reasoning each item supports is stated in the main text; what follows is the raw material needed to re-run or re-check it.

#	condition	constant (src/environment/singlecombat_shoot_task.py)
C1	missiles remaining > 0	NUM_MISSILES = 4
C1b	steps since last launch $\geq$ MIN_ATTACK_INTERVAL	30 steps = 6 s at 5 Hz
C2	target alive	—
C3	ATA < MAX_ATTACK_ANGLE	15.0 deg
C4	MIN_ATTACK_DISTANCE $\leq$ range $\leq$ 3000 + 5000·(AA/180)	1500.0 m
C5	closure < 0 (closing on the target)	—

axis	environment permits	expert additionally requires	relation
ATA	< 15.0 deg	< ATA_STRICT = 10.0 deg	strictly tighter
closure	< 0 m/s	< -CLOSURE_MIN = -5.0 m/s	strictly tighter
range	1500 m $\leq$ r_max(AA)	DLZ depth in [0.25, 0.75], where depth = (range - 1500)/(r_max - 1500)	strictly tighter and two-sided

arm	rule
asap	fire at the first legal step (the cooldown is already inside the mask), then whenever legal
delay_30	fire at the first legal step $\geq$ first-legal + 30
delay_60	fire at the first legal step $\geq$ first-legal + 60
dlz_mid	fire only when DLZ depth $\in$ [0.4, 0.6]
dlz_deep	fire only when DLZ depth $\geq$ 0.6
interval_100	fire ASAP, then at most every 100 steps while legal
bc	the inherited BC launch head (the reference arm)

4.1 E.1 The environment-lifecycle confound — affected scripts and artifacts

\S4.4 states the confound and why the reused-env numbers are excluded. The specifics: both paths import the same rule functions (hdg_label / spd_label / fire_desired) at CMD_SPEED = 280, and BaseEnv.__init__ already installs SafetyInterceptor(PIDFlightController()), so neither the rule nor the controller explains the disagreement. The reused-env path is generate_shoot_rule_expert.py --validate, which builds **one** BaseEnv and reuses it for every episode; the fresh-env path is used by eval_paired_bc_vs_expert.py, eval_bc_1v1.py, fire_oracle_audit.py and eval_asap_baseline.py, each of which builds a new environment per episode. Affected artifacts are renamed EXCLUDED_reused_env_expert_*.json, and collect_matrix.py skips anything marked EXCLUDED_. Gate: tests/check_expert_path_consistency.sh.

4.2 E.2 Corroborating CLR: timestamps and the gate on disallowed steps

\S3.2 quotes the two corroborating values (expert 5.96%, BC 7.37%) and their geometry caveat; the exact provenance is:

- data/expert/shoot_rule_expert.npz, 200 episodes, **mtime 2026-09-13 20:22** — this *predates* commit fb48155 (2026-09-14 09:06), which changed the heading-bias range from $U(30, 60)$ to $U(0, 60)$. Its 5.96% is therefore an **old-geometry** measurement.
- results/health_check/fire_hesitancy.json, 200 rollouts, generated **2026-09-16**, i.e. after the change — so the 7.37% beside it is $U(0, 60)$. The two are **geometry-mixed** and are never averaged or presented as a matched pair.

4.3 E.3 Scope reading for the CLR claim

Neither CLR figure alone proves suboptimality: a low CLR diagnoses restriction only, and whether that restriction is *costly* is established separately (\S4.3, \S4.4). The instrumentation and stability cross-checks are in \S4.2.

4.4 E.4 Compute environment (moved from \S4.1)

Single RTX 3060 Laptop GPU (6 GB, cu126, driver 528.79). Closed-loop evaluation: BC $\approx$ 6.7–13.0 s/episode, SPC $\approx$ 10 s/episode, so one n=400 arm pair is $\approx$ 1.7 h; rule-expert rollout $\approx$ 6.3 s/episode. All runs execute in WSL2 Ubuntu 22.04 with the project interpreter (/home/sean/miniconda3/envs/marl_env/bin/python); reproduction commands are in Appendix D.

4.5 E.5 Ablation A5 — full training and screening detail (moved from \S4.7)

- **Implementation.** scripts/distill_fire_asap.py --random-fire-head: SPC's freeze pattern is kept exactly (encoder and every maneuver head frozen at θ_{BC}) and only the fire head is discarded and rebuilt from nn.Linear.reset_parameters() — the constructor's own initialisation, so it is a genuine random init (weight std 0.049, bias std 0.044) rather than a hand-rolled approximation of one. Mutually exclusive with --full-network (A5 and A6 are different ablations).
- **Training.** 300 rollout episodes (392091 transitions, 11976 allowed steps; train/val split 9578/2398 allowed steps). The random fire head reaches fire-CE 0.0413 with train recall 100.00% and validation recall 100.00% **in its first epoch** (of 15) and triggers the $\geq$ 99.5% early stop. Identity gates: hdg/spd logits diff = 0.00e+00 — *exactly* zero, not

Quantity	Value	Basis
Expert fire_desired on disallowed steps	4.03% (10336 / 256357) — the gate would fire outside the window too; the mask (mostly cooldown) holds it back	data/expert/shoot_rule_expert.npz

Identity measurement	SPC (fire head only)	A6 (full network)
heading/speed logit max-diff vs θ_{BC}	< 1e-9	8.51
heading/speed action sequences identical	yes	no
launch-mask agreement	yes	yes
validation allowed-window accuracy	≥ 0.995	0.9967

Arm (400 seeds, d=0)	Kill rate	CLR	vs SPC (paired)
BC (inherited launch policy)	46.50%	7.26%	—
A6 — full-network fine-tune	77.50% (310/400)	99.67% (1508/1513)	-13.25 pp , discordant 67 (60 SPC / 7 A6), exact McNemar p = 1.33e-11
SPC — launch head only	90.75% (363/400)	100.00%	—

merely under tolerance — with identical action sequences and VERDICT: PASS.

- **Primary protocol.** 90.75% (363/400) kills, CLR 99.94% (1560/1561) — the CLR denominator differs from SPC's by exactly one legal step (1561 vs 1560), the only trace of the different initialisation anywhere in the evaluation.
- **60-seed screening** (same seed set as A6b; ± 10 pp small-sample spread applies): A5 **95.0%** (id) / **86.7%** (dist2_3k) at 100% window utilisation, bit-for-bit equal to the mask-permissive rule oracle on the same seeds; BC 40.0% / 15.0%. The screening set is kept because it carries the third arm.

4.6 E.6 Ablation A6 — protocol, identity measurements and results (moved from \S4.7)

Protocol. Identical objective, data, loss and initialisation as SPC; the only change is that every parameter is trainable rather than the launch head alone, at a deliberately **more generous** budget (—epochs 40 —lr 3e-4 versus —epochs 15 —lr 1e-2), so that a negative result cannot be attributed to under-training.

Identity measurements — the maneuver is untouched under SPC and demonstrably changed under A6:

Kill rate and CLR at the primary protocol (d=0, U(0,60), the same 400 seeds as Table 4):

Full-network fine-tuning recovers most of the gap (46.50% $\rightarrow$ 77.50%) but remains significantly worse than the localized intervention on identical seeds, and pays for it with an 8.51-logit change to the maneuver.

4.7 E.7 Ablation screening set — the mask-permissive third arm (moved from \S4.7)

The ablation script's own 60-seed evaluation (geometry U(0,60), deterministic masked argmax) runs the id and dist2_3k cells side by side and adds the **mask-permissive rule oracle**, which the 400-seed runs do not:

A6 also loses to a rule that simply fires whenever the mask permits (81.7% vs 95.0%). The set is **60 seeds** and carries roughly ± 10 pp of small-sample spread — the same seeds give BC 40.0% where n=400 gives 46.50% — so these figures are quoted only for the third arm and never alongside the 400-seed numbers as one comparison.

4.8 E.8 Ablation tracking table (project bookkeeping)

4.9 E.9 The E2 rule-oracle comparison — identities and limitations (moved from \S4.4)

eval_asap_baseline.py had no —start-seed, so the oracle could only ever run seeds 0..N-1 and could not be matched to a trained policy's evaluation. Patched, then run on the same seed set:

Identities verified from run_meta: the oracle arm loaded sha256 aad05b45\ldots (BC round1 — the *frozen maneuver* source, correct), SPC loaded 36d79bd9\ldots.

One limitation. eval_asap_baseline.py writes aggregates only, with no per-episode detail, so the identity above follows from construction rather than from a seed-by-seed match. Emitting per-episode records would make it mechanically checkable.

5 ARTIFACT PROVENANCE RECORDS

The records below were collected out of the main text, where they interrupted the argument. Each states which artifact backs which claim.

Citation status (re-checked 2026-09-18 against the live record, not from memory). Verified: Pomerleau (NIPS'88, pp. 305-313; presented 1988, MIT Press 1989 - the year is now confirmed, not inferred), Ross/Gordon/Bagnell (AISTATS 2011), Bojarski (arXiv:1604.07316), Brown et al. (ICML 2019 pp. 783-792), Huo/Wang/Xu (AAAI 2023 pp. 7953-7961), Beliaev/Pedarsani (arXiv:2402.01886), Li Z. et al. (NeurIPS 2023), Parisotto/Ba/Salakhutdinov (ICLR 2016, arXiv:1511.06342), McGrew et al. (JGCD 33(5):1641-1654), Pope et al. (ICUAS 2021 pp. 275-284), Yang et al. (IEEE Access 8:363-378), Li L. et al. (Neurocomputing 584:127591, 2024), Li S. et al.

Arm (60 seeds, same seed set)	id cell	dist2_3k cell	window utilisation (id)
BC (inherited launch policy)	40.0%	15.0%	7.2%
A6 — full-network fine-tune	81.7%	66.7%	98.3%
mask-permissive rule oracle	95.0%	86.7%	100%

#	Ablation	Status	Purpose
A1	Fire-policy oracle enumeration on frozen trajectory (\S4.3)	done (E1 regenerated, tracked)	establishes that the inherited launch policy is sub-optimal, maneuver held fixed
A2	SPC (learned head) vs ASAP rule oracle (\S4.4)	done — exact match on one seed set (90.75% both); completeness check, not independent validation	SPC captures 100% of the oracle gap
A3	difficulty_level = 0.3 (\S4.6)	done — +48.50 pp, 194:0, p=7.97e-59	interactive/reacting opponent
A4	Heading-shift robustness (\S4.5)	done (n=400)	benchmark sensitivity
A5	Fire head from random init, encoder frozen (\S4.7, E.5)	done 2026-09-18 (400-seed primary protocol)	warm-start dependence
A6	Full-network fine-tune on the same objective (\S4.7, E.6)	done 2026-09-18 (400-seed primary protocol)	is freezing merely engineering convenience?
A6b	60-seed screening incl. the mask-permissive arm (E.7)	done, secondary	third-arm comparison only
A7	CLR for other binary decisions	conceptual	generality of the metric (future work)

Arm	Kill	Launches/ep	Hit	lost	Termination reasons
Rule oracle (asap: frozen BC maneuver + fire on every legal step)	363/400 = 90.75%	3.90	1.0000	0	target_killed: 363, timeout: 37
SPC	363/400 = 90.75%	3.90	1.0000	0	target_killed: 363, timeout: 37

(ACM TAAS 21(1):6, 2026), **Rusu et al. (ICLR 2016 conference track, arXiv:1511.06295)**, **Hinton/Vinyals/Dean (NIPS 2014 Deep Learning Workshop, arXiv:1503.02531 - a workshop paper, so it has no page numbers and none are invented here)**, **LoRA (Hu et al., ICLR 2022, arXiv:2106.09685)**, **ROME (Meng, Bau, Andonian & Belinkov, NeurIPS 2022, Advances in NeurIPS 35, pp. 17359-17372)**. **Nothing on this list is now unconfirmed.**

Provenance. The measured column comes from results/shoot_eval/E1_fire_oracle_dist2_3k_s60.json, which is version-controlled and carries run_meta with the cell configuration, geometry, difficulty, seed range and the frozen checkpoint's SHA-256. The 2026-08-06 artifact no longer exists; the italic column is all that survived of it,

which is why the regenerated file is tracked rather than regenerated on demand.

Artifact status. E7 is **verified against live artifacts** (results/shoot_eval/E7_bc_round1, spc_d0_n400_s20000.json, E7_paired_bc_vs_spc_d0_n400.json).

Artifact status: verified against live artifacts — results/shoot_eval/E5_bc_round1, spc_d03_n400_s20000.json, E5_paired_bc_vs_spc_d03_n400.json, and the two CLR runs E3_paired_bc_vs_expert_d0_n400_s20000_v3, d03_n400_s20000_v2_clr.json. The expert arm is no longer "running": it is the fresh-env paired arm of E3 at both difficulties, and it is what Table 4 reports.